VECTOR

J1939 Port Interface Catalog

Technical Reference

Technical Documentation

Version 1.0

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Document Information

History

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Reference Documents

No.	Source	Title	Version
[1]	Vector	J1939 Port Interface Catalog (AR4-550) SRS and SDS	1.0
[2]	Vector	Technical Reference Vector Legacy Converter	1.4.2
[3]	Autosar	System Template, R4.0 Rev 3	4.2.0
[4]	Autosar	System Template, R4.1 Rev 2	4.4.0



Caution
We have configured the programs in accordance with your specifications in the questionnaire. Whereas the programs do support other configurations than the one specified in your questionnaire, Vector´s release of the programs delivered to your company is expressly restricted to the configuration you have specified in the questionnaire.

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1 Introduction

This document describes the generation of J1939 Port Interface Catalogs from DBC files using the Vector Legacy Converter (VLC). The J1939 Port Interface Catalogs are imported by the DaVinci Developer, and they serve as templates for creating port prototypes on application components which then can be automatically connected to J1939 signals derived from the same DBC files.

2 Vector Legacy Converter Functionality

For technical reasons the J1939 Port Interface Catalog generator has been implemented as part of the VLC. The calling conventions and options of the VLC for creating J1939 Port Interface Catalogs are specified with the following help text:

```
Usage: LegacyDb2SystemDescrConverter [options] <file> [<file> ...]

Options:
  -r, --release <40|41>      AUTOSAR release
  -p, --ports                Create J1939 port interface catalog
  -o, --output <file>        Output file

Parameters:
  <file>                     Input file (*.dbc)
```

Table 2-1 Vector Legacy Converter arguments.

The parameters `-r` and `-p` are mandatory when creating a J1939 Port Interface Catalog. The release parameter defines whether the output file format is AUTOSAR 4.0.3 (`-r 40`) or AUTOSAR 4.1.2 (`-r 41`). A typical VLC call would be:

```
LegacyDb2SystemDescrConverter -r 41 -p -o J1939Ports D:\Data\j1939.dbc
```

The VLC takes all J1939 application messages and creates `PORT-PROTOTYPE-BLUEPRINT` and `SENDER-RECEIVER-INTERFACE` elements for them, while J1939 protocol messages will be ignored. In addition, for each J1939 application signal a `VARIABLE-DATA-PROTOTYPE` element is created. These J1939 elements together form the J1939 Port Interface Catalog of a DBC file. The derivation of J1939 elements is exemplarily shown for a DBC message named “MsgJ1939” which contains an unsigned 3 bit signal named “SigJ1939”:

AUTOSAR Element	Name
PORT-PROTOTYPE-BLUEPRINT	PP_MsgJ1939
SENDER-RECEIVER-INTERFACE	PI_MsgJ1939
VARIABLE-DATA-PROTOTYPE	PI_MsgJ1939/DE_SigJ1939
DATA-CONSTR	DC_0x0_0x7
IMPLEMENTATION-DATA-TYPE	uint8
DATA-CONSTR	uint8

Table 2-2 J1939 Port Interface Catalog elements.

3 Glossary and Abbreviations

3.1 Glossary

Term	Description
J1939	J1939 is the vehicle bus recommended practice used for communication and diagnostics among vehicle components.
PORT-PROTOTYPE-BLUEPRINT	This meta-class represents the ability to express a blueprint of a PortPrototype by referring to a particular PortInterface. This blueprint can then be used as a guidance to create particular PortPrototypes which are defined according to this blueprint. By this it is possible to standardize application interfaces without the need to also standardize software-components with PortPrototypes typed by the standardized PortInterfaces.
SENDER-RECEIVER-INTERFACE	A sender/receiver interface declares a number of data elements to be sent and received.
VARIABLE-DATA-PROTOTYPE	A VariableDataPrototype is used to contain values in an ECU application.

3.2 Abbreviations

Abbreviation	Description
AUTOSAR	Automotive Open System Architecture
MICROSAR	Microcontroller Open System Architecture (Vector AUTOSAR solution)
VLC	Vector Legacy Converter

4 Contact

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